A Multivariable Industrial Plant: the Boiler–Turbine Unit

Multivariable Control, Homework 1

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Task

Introduce an industrial multivariable plant and identify its inputs and outputs.

What I wrote

Following Maffezzoni, *Boiler–turbine dynamics in power-plant control* (Control Eng. Practice, 1997), I described the boiler–turbine unit of a fossil-fuel power plant as a strongly coupled MIMO system built from three dynamics:

- **Evaporator pressure.** Set by the balance between heat input (mainly from fuel flow) and the energy carried away by steam.
- **Drum level.** Set mainly by the mass balance between feedwater and steam flow, and also disturbed by heat input and pressure swings.
- **Electrical power.** Proportional to high-pressure turbine steam flow, plus a delayed contribution from the reheater.

Inputs: fuel flow, air flow, turbine admission (control-valve opening), feedwater flow, recirculated-gas flow, and superheater and reheater spray flows.

Outputs: electrical power, turbine inlet pressure, drum level, turbine inlet temperature, reheater outlet temperature, and air-to-fuel ratio.

Fuel flow changes pressure, which changes drum level, and heat input affects both power and level. Because of this coupling, independent single loops are not enough; the plant needs coordinated multivariable control.